

Husna Manalai

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Education

Bard College, B.A. in Computer Science (Mathematics minor) Sept 2022 – May 2026

- **GPA:** 3.53/4.0
- **Relevant Coursework:** Object-Oriented Programming, Data Structures, Algorithms, Discrete Math, Introduction to AI, Databases, Software Development, Machine Learning, Linear Algebra, Calculus I, Calculus II, Design of Programming Languages, Vector Calculus, Statistics for Computing, Senior Project

Research Experience

Research Intern, Bard Summer Research Institute – Red Hook, NY June 2025 – Aug 2025

- Continued research on the mathematics of the card game *Quads*.
- Proposed and proved the *6-Theorem of Quads*, establishing that in a specific 8-card configuration without quads, two 3-card subsets determine a quad that also applies to the remaining cards.
- Built *Daily Quads*, a JavaScript webpage that generates a new puzzle each day.

Research Intern, Bard Summer Research Institute – Red Hook, NY May 2024 – Sept 2024

- Researched the card game called *Quads* using finite geometry and combinatorics.
- Developed Python code to compute the probability of a quad in a k -card layout more efficiently.
- Collaborated on a multiplayer *Quads* game in Unity enabling online play for two players.
- Designed *Kitten Kaboodle*, a new math game inspired by finite projective geometry and the game of SET.

ASC-Lab Research Intern, Bard Summer Research Institute – Red Hook, NY May 2023 – Sept 2023

- Generated and simplified quandle expressions arising from knot diagrams, applying right-cancellation (right-reduction) properties to prepare test cases for unification algorithms.
- Worked with knot colorings by constructing crossing equations and visualizing nontrivial quandle colorings using OmniGraffle.
- Developed and used Prolog (backtracking) and Python scripts to generate quandle expressions and search for valid colorings in Sharac- j quandles.
- Examined the computational complexity behavior of quandle coloring algorithms with respect to knot crossings and quandle size.

Teaching Experience

Math Tutor, Bard College – Red Hook, NY Sept 2023 – May 2025

- Tutored Bard students one-on-one in introductory and intermediate Math courses: Algebra, Pre-Calculus, Calculus I and Calculus II.
- Tutored middle school and high school students through the Bard Math Circle in Algebra I,

Algebra II, and Pre-Calc.

Computer Science Tutor, Bard College – Red Hook, NY Sept 2023 – May 2025

- Tutored Bard students one-on-one in introductory and intermediate computer science courses: Object-Oriented Programming, Data Structures, Discrete Math, Introduction to AI.
- Tutored middle school and high school students through the Bard Math Circle in Processing Programming Language and Python.

Volunteer Math Tutor, Interfaith Rise – Highland Park, NJ Dec 2021 – Jan 2022

- Tutored math for Afghan refugee women through a community education program.
- Covered foundational topics including Arithmetic, Algebra, and Pre-calculus.

Professional Experience

Trustee Leader Scholar (TLS) Leader, Bard College – Red Hook, NY Sept 2025 – Present

- Lead a math outreach program, *Mathfinity*, as part of the Trustee Leader Scholar program at Bard College.
- Design and facilitate interactive math activities using puzzles and games for students and community members.
- Organize and host regular math events for homeschool students.

Data and Technology Intern, Third Act – Brooklyn, NY Sept 2024 – Dec 2024

- Led weekly data cleaning and de-duplication efforts on large datasets within the EveryAction CRM, improving data accuracy and outreach efficiency.
- Maintained Working Groups lists, improving team collaboration and communication efficiency.
- Wrote Google Apps Script to format weekly ActBlue contributions, streamlining donation tracking.

Projects

Daily Quads Puzzle Webpage

- Built an interactive web application that generates a new Quads puzzle each day.
- Implemented puzzle logic, user interaction, and daily state updates in JavaScript.
- Link: husnamanalai.com/quads
- Tools: JavaScript, HTML, CSS.

Predictive Analytics for Bike Rental Demand

- Built and compared two machine learning models: a Fully Connected Neural Network (FCNet) for regression and a Recurrent Neural Network (RNN) for time-series prediction.
- Evaluated models using MSE and R^2 and analyzed temporal and seasonal patterns.
- Tools: Python, TensorFlow, NumPy.

Smart Recipe Recommendation Engine

- Designed a recipe recommendation platform using cosine similarity to personalize suggestions.
- Implemented the model in Python using pandas and scikit-learn for feature extraction and similarity scoring.

Publications, Talks and Poster Presentations

Joint Mathematics Meetings (JMM 2026), Talk

Jan 6, 2026

- *Quads: Shapes and Daily Puzzle*
- Structure and shapes of quads, and conjectures about the Daily Quads Puzzle.

MOVES Conference 2025, Talk

Aug 12, 2025

- *Quad Shapes and Quad Packing*
- Shape of quads in 8-card configurations and how many quads we can pack in different n -card configurations.

NYC Summer Math Day at Jane Street, Poster Presentation

July 21, 2025

- *Flats and Caps in Quads*
- Finite geometric interpretation of quad configurations.

Bard Summer Research Institute Poster Session, Poster Presentation

Oct 24, 2025

- *Flats and Caps in Quads*
- Theoretical and computational results from summer research.

Technical Skills and Languages

- **Technical:** Java, Python, SQL, HTML, CSS, JavaScript, Prolog, Haskell, AWS Services, Google Apps Script
- **Languages:** English, Persian, Pashto, Turkish